

Track 2.2 Document Processing

Module Name: Research Paper Document Processing Engine

Objective

Convert uploaded research papers into machine-readable text.

Module Architecture

PDF/DOCX Upload



File Validation



Metadata Extraction



Text Extraction



Structured Output

Submodule 2.2.1: Document Upload Service

APIs

POST /upload

GET /documents

DELETE /documents/{id}

Validation

Allowed:

pdf

docx

txt

Check:

- size limit
 - duplicate files
 - corrupted files
-

Submodule 2.2.2: PDF Extraction Engine

Libraries

1. PyMuPDF
2. pdfplumber
3. PyPDF2

Workflow

PDF

↓

Pages

↓

Text

Output Like:-

```
{  
  "File id": "...."  
  "heading": "...."  
  "page": 1,  
  "text": "..."  
}
```

Submodule 2.2.3: DOCX Extraction Engine

Library

python-docx

Workflow

DOCX

↓

Paragraphs

↓

Text

Submodule 2.2.4: Metadata Extraction

Extract

Basic Metadata

Title

Author

Page Count

File Size

Advanced Metadata

Keywords

DOI

Publication Year

(if available)

Output Like:

```
{  
  "title": "...",  
  "authors": "...",  
  "year": "..."  
}
```

Submodule 2.2.5: Document Repository

Database Table

documents

Field	Type
document_id	UUID
title	TEXT
authors	TEXT
path	TEXT
upload_date	DATETIME

Deliverable

Document catalog.

Submodule 2.2.6: Text Storage Layer

Output Structure

```
{  
  "document_id":"123",  
  "page":1,  
  "text":"..."  
}
```

Storage

documents/
raw_text/
metadata/

Submodule 2.2.7: Quality Validation

Checks

Extraction Success

Text length > 0

Corrupted Pages

Identify blank pages.

Scanned PDFs

Detect image-only pages.